# YAKEEN

# NEET 2.0 2027

Physics | Lecture - 01

## Vector

By- Manish Raj (MR Sir)



Todays Goal

# Questions on log.

# feel of vector

- Basic calculation → Recorded (Today) (PPT Revision)
- Trigonometry ✓
- Phasor diagram ✓
- A/P , GP ✓
- graph - straight & all ✓
- slope variation ✓
- differentiation ✓
  maxima - minr
- Integration ✓
- log ✓
- Binomial ✓
- Quadratic eqr ✓

⇒ H/W

- $\log_8 16 =$ $\log_{(2)^3} 2^4 = \frac{4}{3}\log 2^2 = \frac{4}{3}$
- $\log_{27} 3 =$ $\log_{(3)^3} 3 = \frac{1}{3}\log 3^3 = \frac{1}{3}$
- $\log_{100} 1000 =$ $\log_{10^2} 10^3 = \frac{3}{2}\log_{10} 10 = \frac{3}{2}$
- $\log_{10} (0.0001) =$ $\log 10^{\left(\frac{1}{10^4}\right)} = \log 10^{10^{-4}} = -4\log_{10} 10 = -4$
- $\text{Log}_{0.01} 10 =$ $\log_{(10^{-2})} 10 = -\frac{1}{2}\log_{10} 10 = -\frac{1}{2}$ ✓

## Question

H/w must do copy

Loudness of sound at a point is 50 dB then find intensity at that point if $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB where $I_0 = 10^{-12}$ w/m²

1. $10^{-7}$ w/m²
2. $10^{-5}$ w/m²
3. $10^{8}$ w/m²
4. $10^{7}$ w/m²

$L = 50\,dB$ ✓

$L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB $\quad I_0 = 10^{-12}$

~~$50\,dB = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB~~

$5 = \log_{10}\left(\frac{I}{I_0}\right)$ → Antilog ⇒ $10^5 = \frac{I}{I_0}$

$I = 10^5 \times I_0$

$= 10^5 \times 10^{-12} = 10^{-7}$

log ke expression ke samen Power hota hai (Base ka)

**Question**

If current flowing through wire is $I = 20\ e^{-\lambda t}$ find the time when current becomes 10 Amp. Where $\lambda$ is 2.303.

$$I = 20\ e^{-\lambda t}$$

$$10 = 20\ e^{-\lambda t}$$

$$\frac{1}{2} = e^{-\lambda t}$$

$$2^{-1} = e^{-\lambda t}$$

taking log both side on base e

$$\log_e 2^{-1} = \log_e e^{(e)^{-\lambda t}}$$

$$-1 \log_e 2 = -\lambda t \log_e e$$

$$t = \frac{\log_e 2}{\lambda} = \frac{2.303 \log_{10} 2}{2.303}$$

$$= \log_{10} 2 = 0.30$$

## Question

Loudness at a point is 16 dB where intensity is $I$ then find loudness at a point where Intensity is $I/4$.

$L = 16$ dB

where $L = 10 \log_{10}\left(\frac{I}{I_0}\right)$ dB

$16\,dB = 10 \log_{10}\left(\frac{I}{I_0}\right) dB$ — (1)

$L' = 10 \log_{10}\left(\frac{I}{4I_0}\right)$ — (II)

Divide karne ki galti mat karna.

(I) − (II)

$16\,dB - L' = 10 \log_{10}\left(\frac{I}{I_0}\right) - 10 \log_{10}\left(\frac{I}{4I_0}\right)$

$16\,dB - L' = 10\left[\log_{10}\left(\frac{I}{I_0} \times \frac{4I_0}{I}\right)\right]$

$\Rightarrow 16 - L' = 10(\log_{10} 4) = 10 \times 0.6 = 6$

$16 - L' = 6 \hookrightarrow L' = 16 - L = 10$ dB

① $\log_e (x)^n = n \log_e x$ ✓

② $\log_{(e)^n} x = \frac{1}{n} \log e^x$

③ $\log_{10} (xy) = \log_{10} x + \log_{10} y$

④ $\log_{10} \left(\frac{x}{y}\right) = \log_{10} x - \log_{10} y$

⑤ $\log_y x = \frac{1}{\log_x y}$

⑥ $\log_e 1 = 0$

⑦ $\log_a a^a = 1$

$\log_{10} 10 = 1$

$\log_e e = 1$

⑧ $\log_y x \cdot \log_x y = 1$

# Binomial theorem

$(a+b)^2 = a^2 + 2ab + b^2$

$(1+x)^2 = 1 + 2x1x + x^2$

If $1 >>>> x$ then $x^2 \rightarrow$ o(negligible)

$$(1+x)^2 = 1+2x$$

$x <<<< 1$

$$(1+x)^4 = 1+4x$$

$$(1+x)^n = 1+nx$$

$1 >>>> x$

$$(1-x)^n = 1-nx$$

$$(1+x)^{-n} = 1-nx$$

Ex $(1.007)^5 = (1+0.007)^5$

$= 1 + 5 \times 0.007$

$= 1 + 0.035$

$= 1.035$ //

MR. BOX

**

$$(\text{Career} + \text{Love})^2 = \text{Career} + 2\times\text{Love}$$

If $\text{Career} >>>> \text{Love}$ (negligible)

(or)

$(1.007)^5 = 1.035$ ✓

$= 1.035\,19$

$(1.0008)^3 = 1.0024$ ✓

$(1.035)^2 = 1.070$ ✓

$$\begin{bmatrix} 1^5 = 1 \\ 2^5 = 32 \\ 3^4 = 81 \end{bmatrix}$$

$(1.006)^4 = 1.024$

$(1.0007)^5 = 1.0035$

$(1.008)^7 = 1.056$

$(1.02)^6 = 1.12$

$$\sqrt{0.99} = (1-0.01)^{1/2} = 1 - \frac{0.010}{2}$$

$$= 1 - 0.005$$

$$= 0.995$$

$$\sqrt{1.0016} = (1+0.0016)^{1/2}$$

$$= 1 + 0.0008$$

$$= 1.0008$$

$$(1.004)^{-2} = 1 + 0.004(-2)$$

$$= 1 - 0.008$$

$$= 0.992$$

$$(1.015)^{1/3} = 1 + 0.015 \times \frac{1}{3}$$

$$= 1 + 0.005$$

$$= 1.005$$

$$\frac{1}{(1.004)^2} = (1.004)^{-2}$$

(Q) Acceleration due to gravity at height h from surface is $g_h = \frac{GM}{(R+H)^2}$ then find expression of g if $h \ll\ll\ll R$.

2 ways

$$g_h = \frac{GM}{(R+H)^2}$$

$$= \frac{GM}{(R+2H)}$$ ✗ Wrong

$$(R+H)^2 = \left[R\left(1+\frac{H}{R}\right)\right]^2$$

$$= R^2\left(1+\frac{H}{R}\right)^2$$

$$g_h = \frac{GM}{R^2\left[1+\frac{H}{R}\right)^2}$$

$$g_h = \frac{GM}{R^2}\left(1+\frac{H}{R}\right)^{-2} = \frac{GM}{R^2}\left(1-\frac{2H}{R}\right)$$ ✓

step1

$(2.01)^3 = 2.03$ ✗

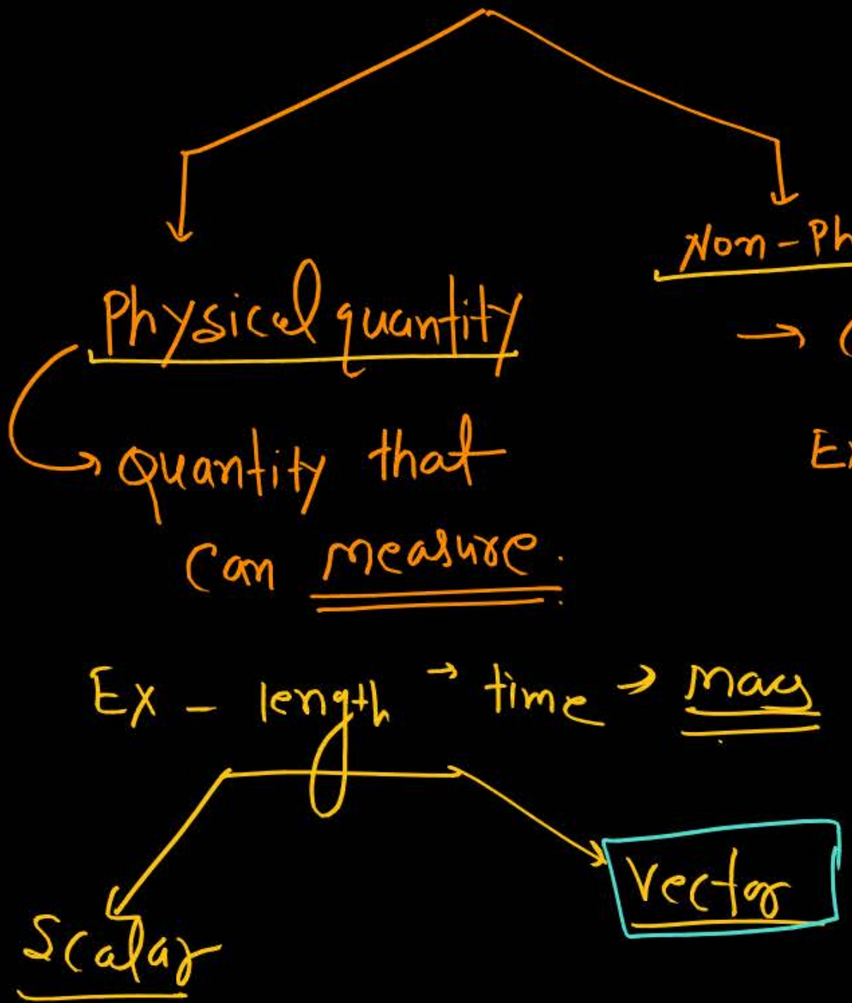
$(2 + 0.01 \times 3) = 2 + 0.03$

$= 2.03$ ✓

$(1+x)^n = 1+nx$

$1 >>>> x$

Vector




Kitna samajh aa raha hai??

(a) 10 Kg

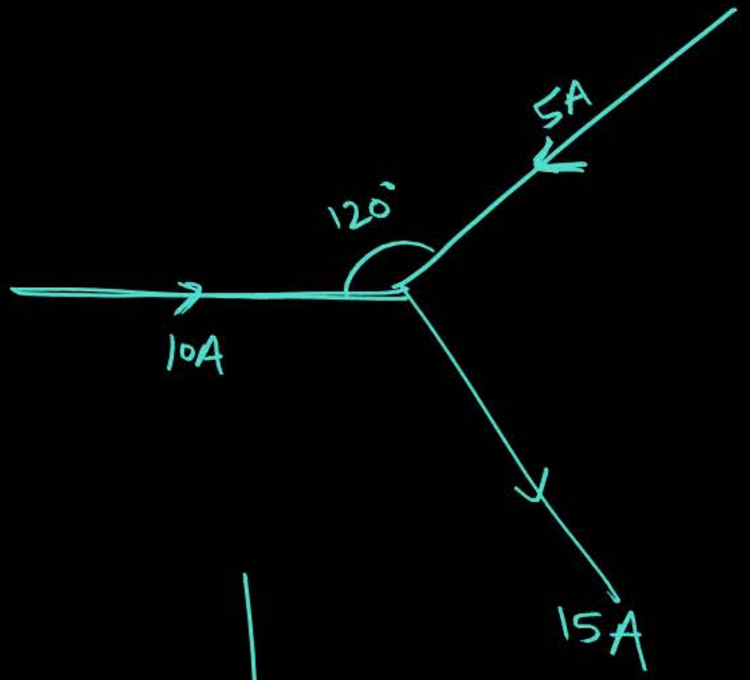
(b) 50 m

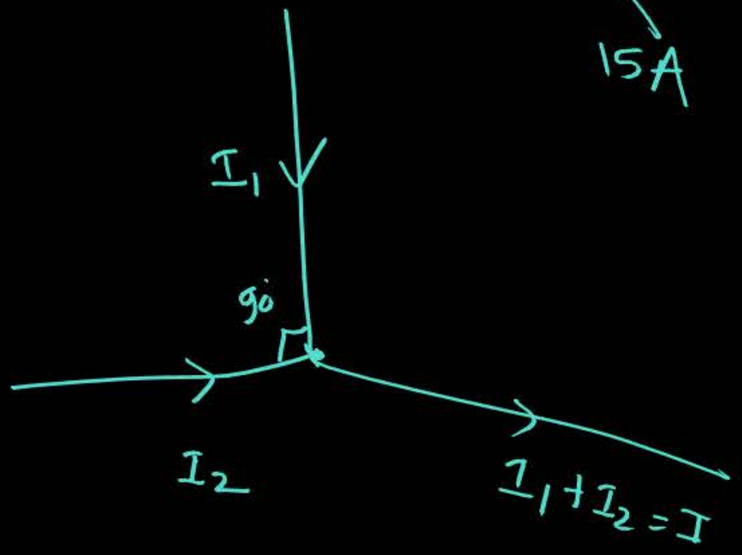
(c) 100 sec

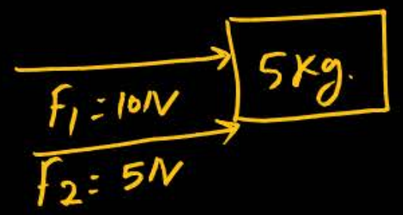
(d) 400 K.

(e) 100.0/0

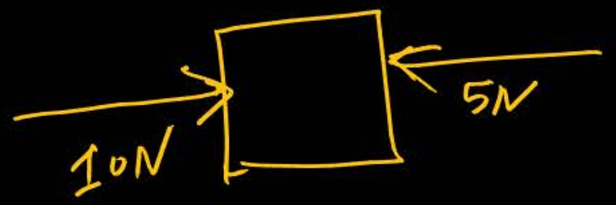
Not copy




$F_{net} = 15N$




$F_{net} = 10 - 5 = \underline{\underline{5N}}$ ✓

# Physical Quantity

## Scalar

- Having only magnitude, does not have direction.
- follow simple algebric addition/sub
- Ex- Mass, Temp, current
  Speed (How fast), distance
  Potential, energy.
  → may be +ve & -ve.
  → In Scale +ve & -ve represent it value [less or great]

$T_1 = +10°C$

$T_2 = -10°C$

$T_2 < T_1$

## Vector

- Having magnitude direction and follow Triangle law of vector addition.
- Ex — force, Velocity, Electric field
  displacment magnetic field.
- for vector +ve & -ve represent direction

$\vec{F_1} = +10N\hat{i}$

$\vec{F_2} = -10N\hat{i}$

equal force with opposite dir

## Mathematical representation

Force = $\vec{F}$

velocity = $\vec{V}$

Electric field = $\vec{E}$

magnitude + direction.

$\vec{A} = A\hat{A}$

$A$ → magnitude

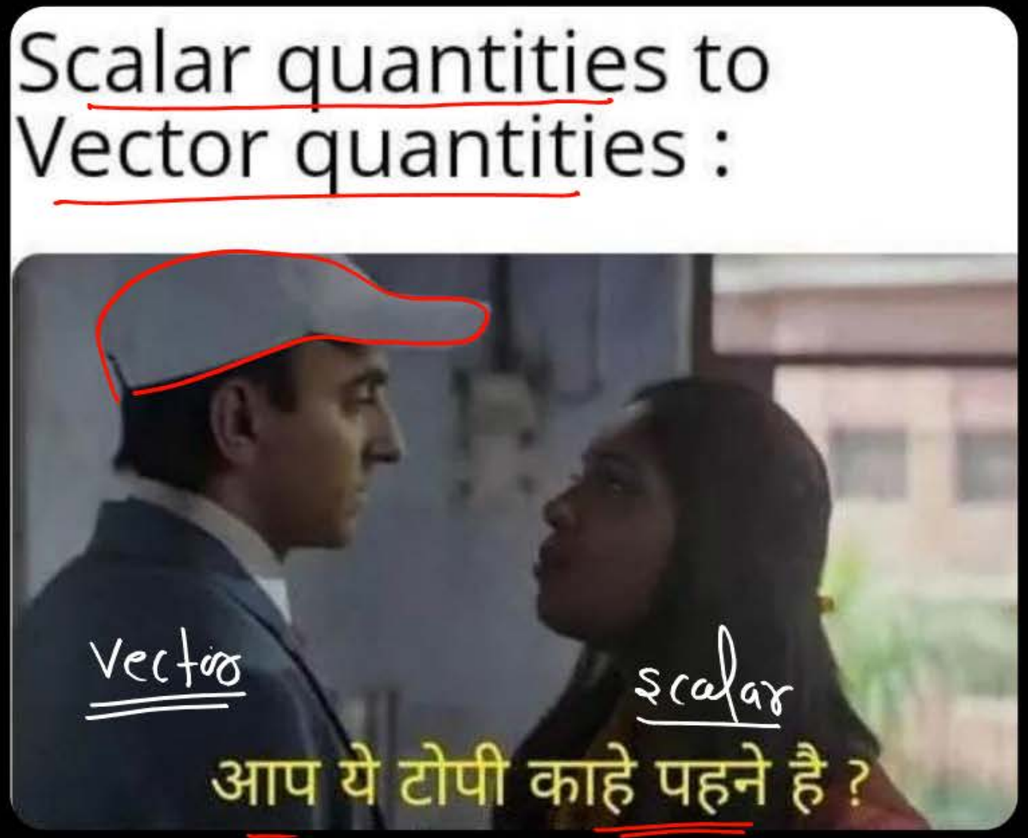
$\hat{A}$ → unit vector for direction.

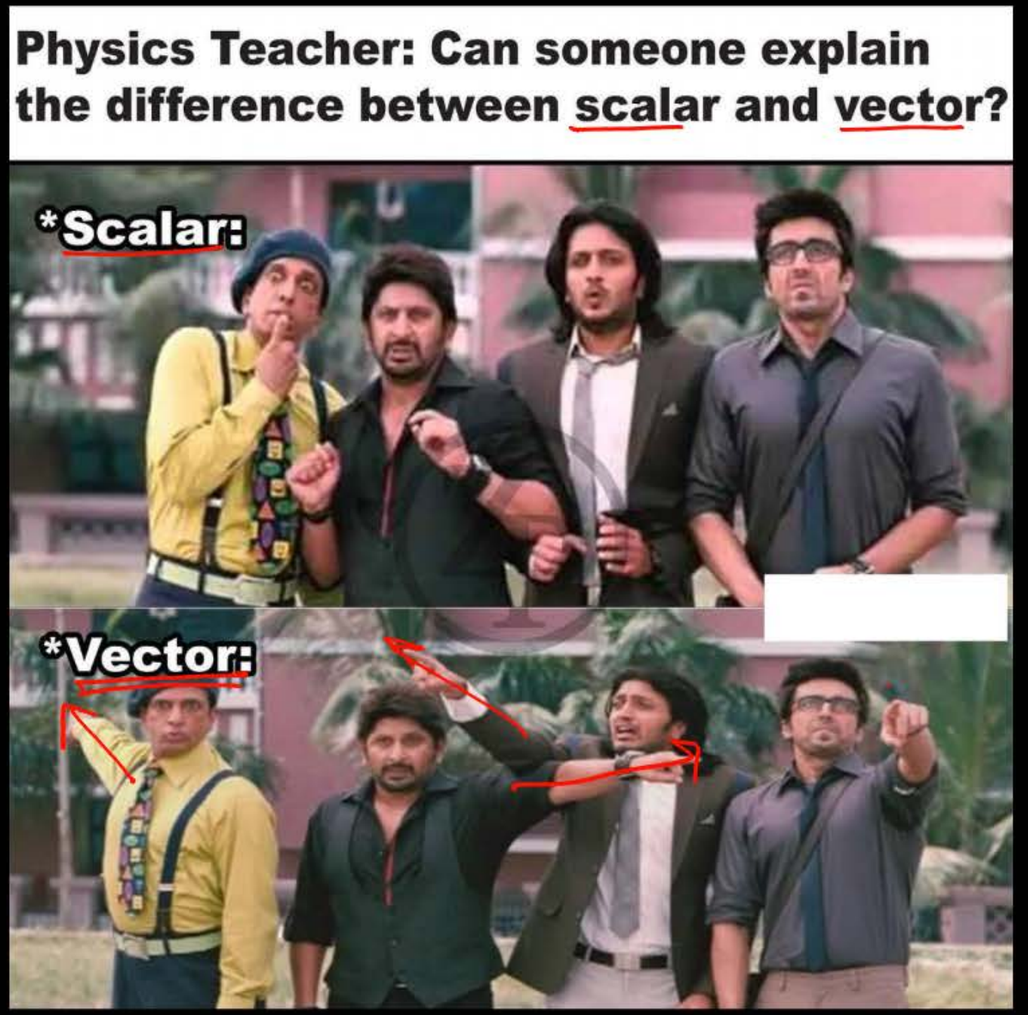
Mass = M

Energy = E

speed = ✓

only magnitude




Physics Teacher: Can someone explain the difference between scalar and vector?
*Scalar:
*Vector:

**Question**

MR Scam

(1) Which of the following is vector?




1. Electromotive force = Potential diff$^{n}$ (scalar) (70%)
2. Surface tension $= \frac{f}{\ell} = \frac{S}{\text{scalar}}$
3. Weight (13%) = gravitational force = mg (downwards)
4. Focal length (f) wrgm = 55kg

## Question

Which of the following representation is correct?

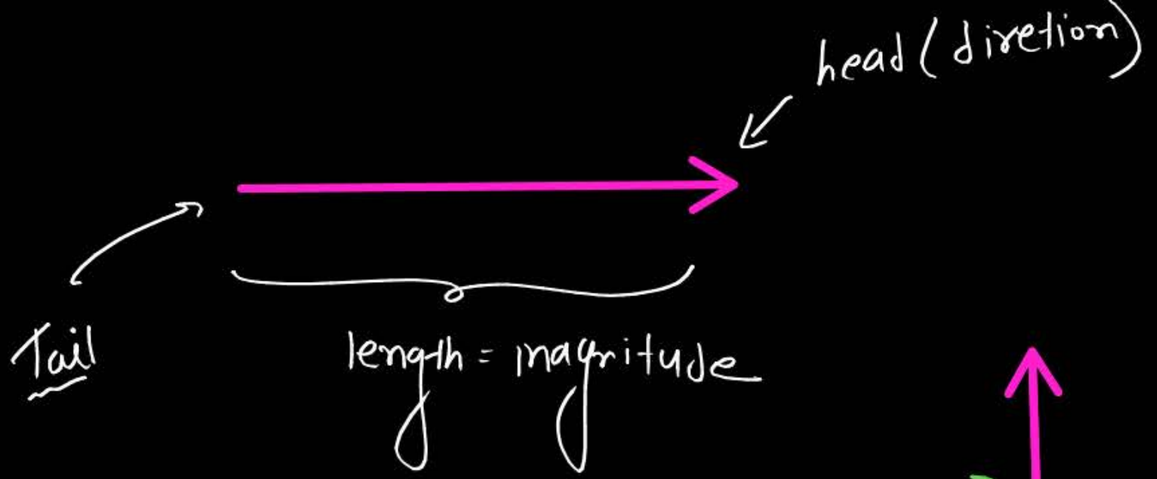
1. Work ($\vec{W}$) (26%)
2. Force (F)
3. Energy (E) (51%)
4. All of these

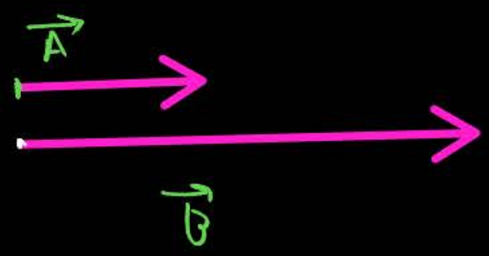
## Question

A. Physical quantity which does not have direction must be scalar.
B. Physical quantity which have direction must be vector.

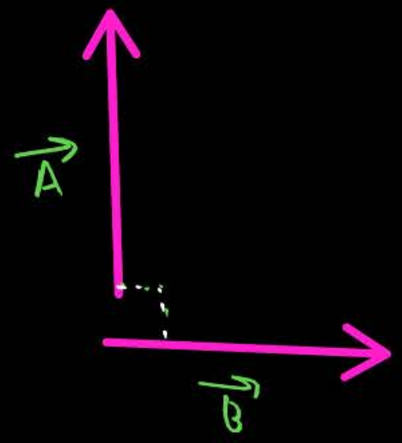
1. Both are true
2. Both are false
3. A true B false
4. A false B true

# graphical representation of vector

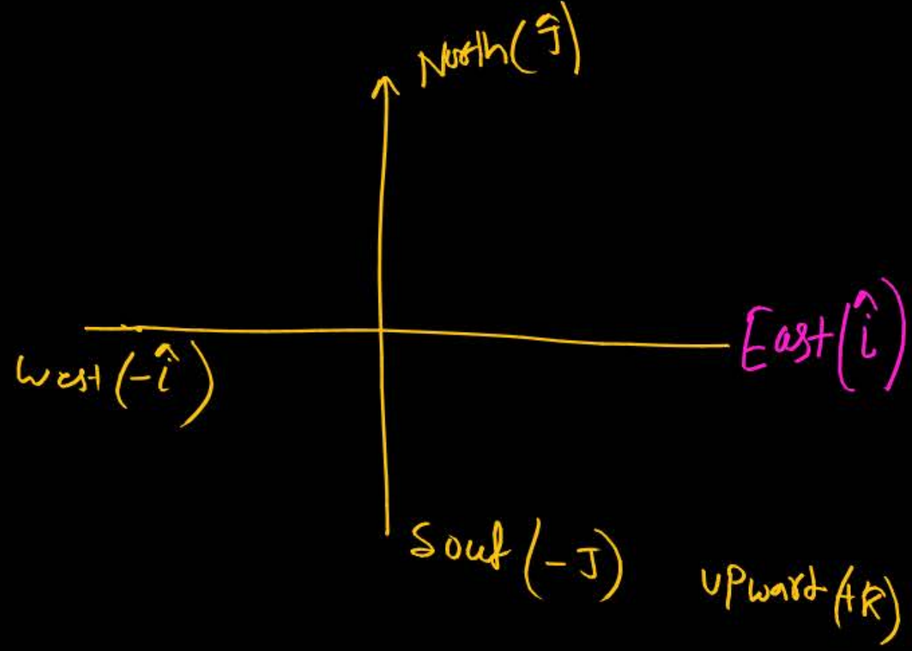



- Same dirn
diffⁿ magnit

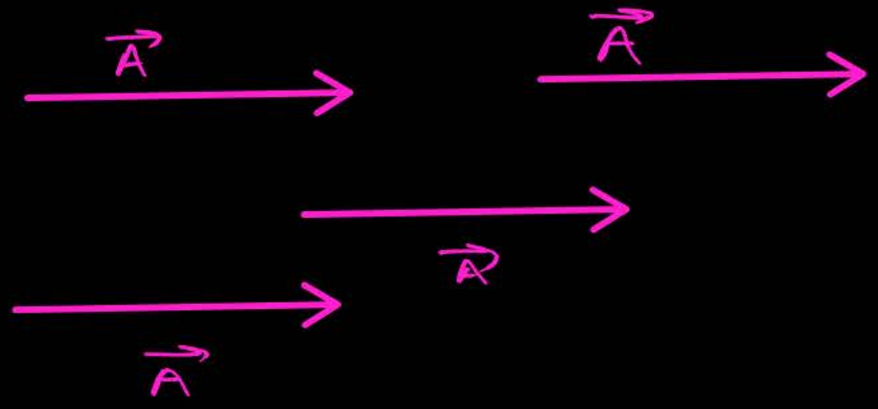



Same magnitud
diffⁿ direction



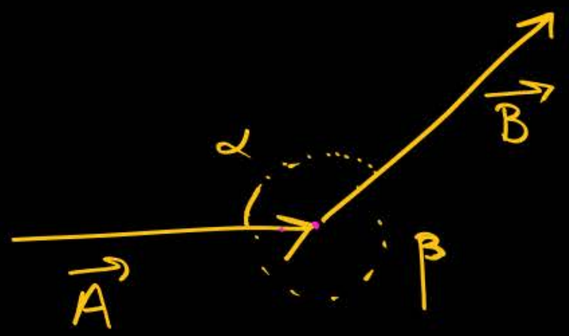
⇒ Vector ko rotate krne se dirn change hoti hai, but vector can be shift parallel to itself keeping its magnitude direction same.

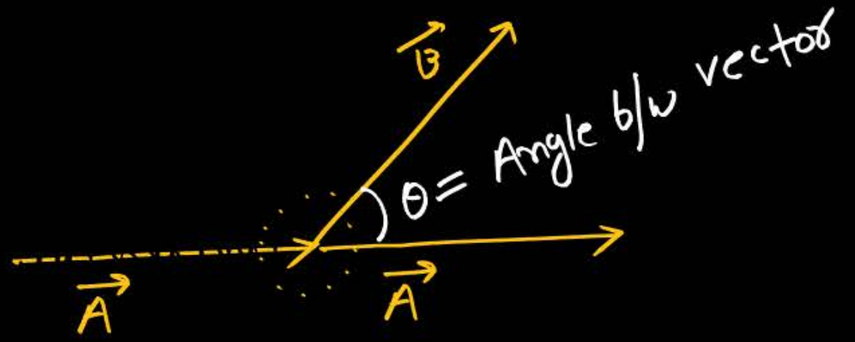



all are same

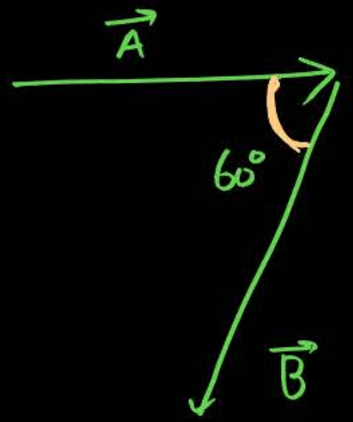
# Angle between vector is a smaller Angle b/w them when they are Placed tail & tail or head & head together #

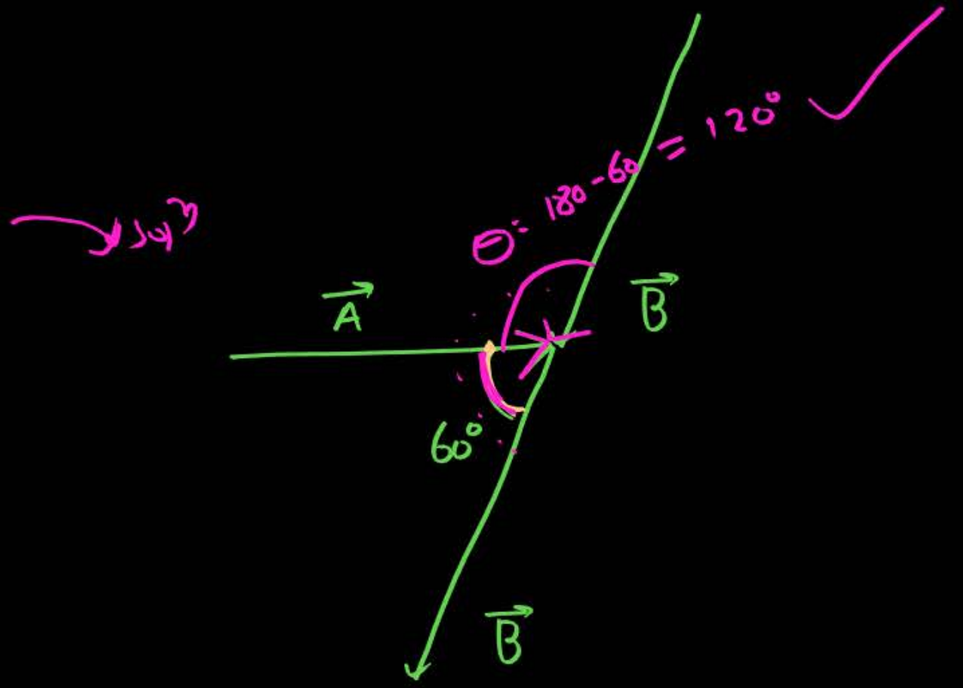



# α & β are not angle b/w $\vec{A}$ & $\vec{B}$

Angle b/w $\vec{A}$ & $\vec{B}$



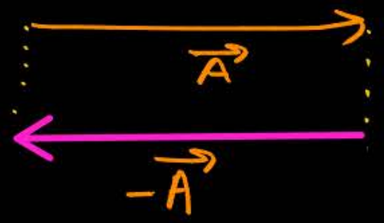



$-$ vector  vector

↓ magnitude same dir$^{n}$ opposite

$$\vec{A} = 10\hat{i}$$

$$-\vec{A} = -10\hat{i}$$



(copy)

② If force $F = 40N\,\hat{j}$ (North) then find Value of $\hat{F}$ is ??

(a) 1N in north ✓

(b) 1 in north (48.1.)

(c) Non-of-these

(d) both (a) & (b)

$\vec{F} = F\,\hat{F}$

$\vec{F}$ → vector

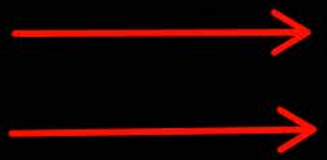
$F$ → magnitude. Have unit

$\hat{F}$ → dirⁿ direction does not have unit.

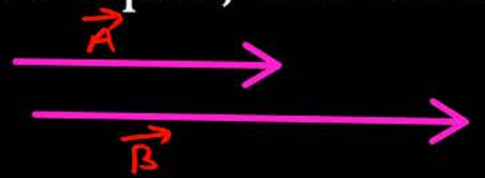
# Types of Vector

Copy

**(A) Equal Vector:** Two vector having same magnitude and same direction of same nature.



**(B) Parallel Vector:** Magnitude may or may not equal, but direction must be same.




# All equal vectors are parallel ✓

# All parallel vector may or may not equal vector ✗

# Types of Vector

**(C) Negative Vector:** Magnitude must be same direction exactly opposite.



**(D) Anti-Parallel Vector:** Magnitude may or may not equal, but direction opposition.



# All –ve vectors are anti-parallel ✓

# All anti-parallel may or may not be negative at each other.



# Types of Vector

**(E) Unit Vector:** A vector having unit magnitude & represent only direction, does not have unit.

**(F) Zero Vector (Null Vector):** A vector having zero magnitude.

Ex. Velocity of car which is at rest. $\vec{v} = 0$

# Types of Vector

**(G) Colinear Vector:** All vector are called colinear when they are along the same line.

**(H) Coplaner Vector:** Vectors are in same plane

# Types of Vector

**(I) Orthogonal Vector (Perpendicular Vector):** Magnitude may or may not same, angle must be 90°.

# Types of Vector

**(J) Concurrent Vector:** Magnitude & direction may or may not be same, but point of action is at same point.

**(K) Axial Vector:** A vector having direction along axis of rotation.

**Ex:** Angle, torque angular velocity, acceleration angular momentum

# Thank

## You